

Welcome to A Catholic Introduction to Artificial Intelligence Class 7! AI in Industry, Healthcare, and Finance

"AI is a witness to truth at best, never the source of truth."

1. Overview

This class explores how AI is being used in major industries such as healthcare, finance, agriculture, transportation, manufacturing, and customer service. Students examine both the benefits and risks of AI-driven systems while continuing the course theme of separating practical reality from hype.

2. Learning Objectives

By the end of this class , students should be able to:

1. Identify several major industries currently using AI.
2. Explain how AI improves efficiency or decision-making in those industries.
3. Describe real-world risks and limitations of industrial AI systems.
4. Recognize how bias, automation, and overreliance on AI can affect society.
5. Evaluate AI systems using ethical and faith-informed reasoning.
6. Research and communicate how AI is applied in a chosen field.

3. Class 6 Review

Class 6 focused on the relationship between artificial intelligence, truth, morality, and ethics from both a technological and Catholic perspective. We explored how AI systems such as large language models generate responses, where their information comes from, and why AI does not “understand” truth in the same way human beings do. The lecture explained that AI predicts patterns in language based on massive amounts of training data gathered from books, articles, websites, and other digital sources. Because of this, AI can produce convincing answers even when the information is inaccurate or misleading.

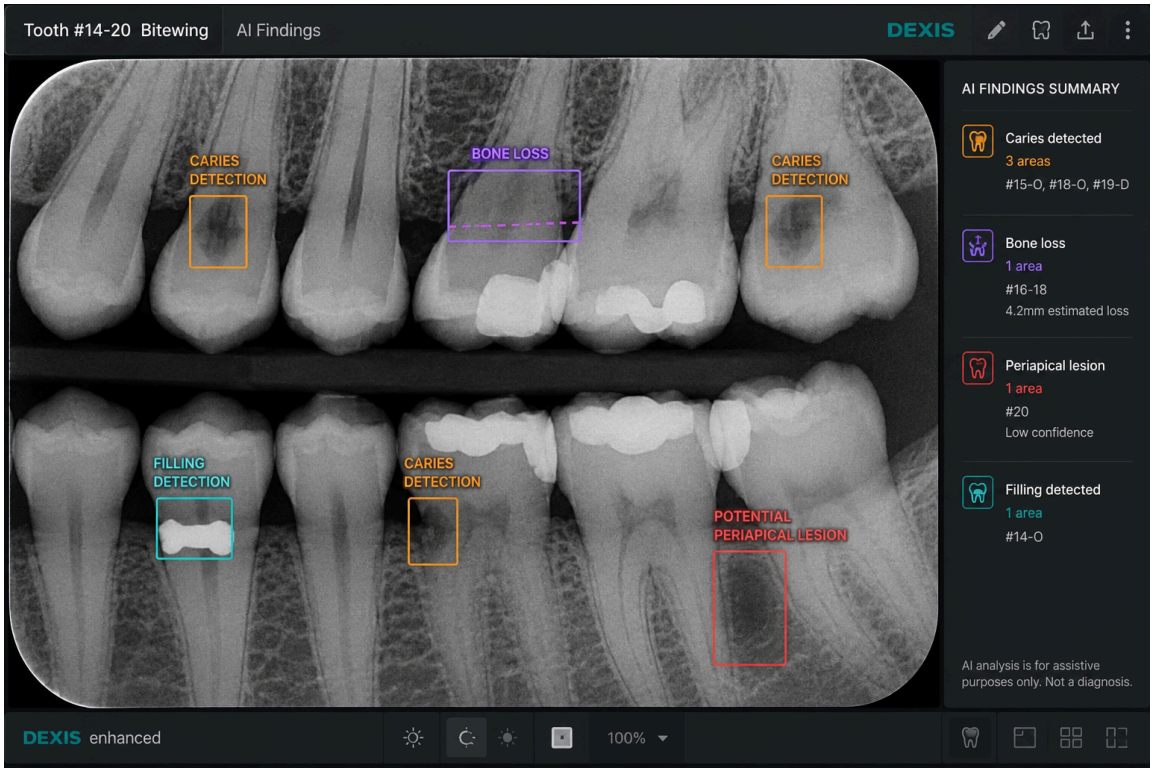
A major theme of the lesson is the concept of truth itself. We examined the difference between factual accuracy, opinion, bias, and objective truth. The lecture discusses how AI systems do not possess reason, conscience, or wisdom. Instead, they statistically generate likely responses based on patterns in their data. This creates important challenges because AI may confidently provide false information, reflect cultural biases found in training data, or present morally questionable content without understanding the consequences. Students are encouraged to think critically about evaluating information rather than blindly trusting AI-generated responses.

From a Catholic perspective, the lecture explored how human dignity, free will, and moral responsibility distinguish people from machines. We learned that human beings are made in the image of God and possess rational souls capable of moral judgment, love, and genuine understanding. AI, by contrast, does not possess consciousness, spirituality, or moral agency. The lesson encourages students to consider how Catholics should approach emerging technologies thoughtfully, ensuring that AI serves humanity rather than replacing authentic human relationships, truth, or moral reasoning.

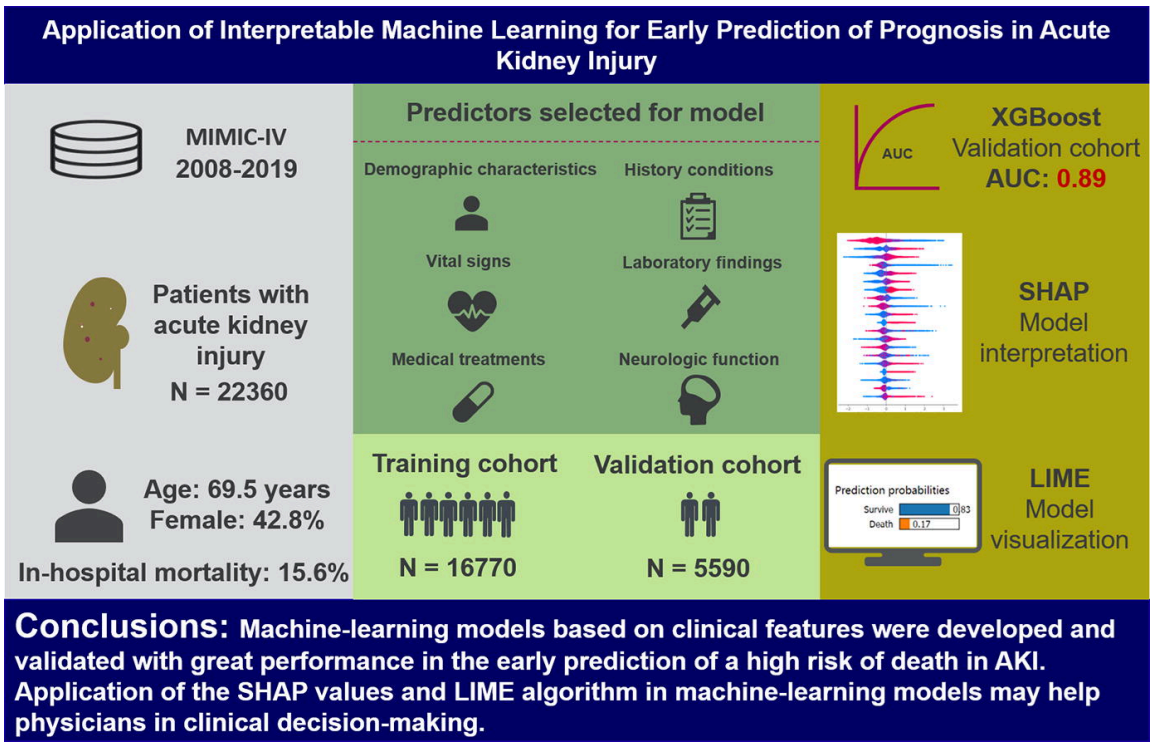
Finally, the lesson encouraged us to reflect on our own use of AI tools. We are challenged to think about when AI can be helpful, when it should be questioned, and how personal responsibility remains essential in an age of rapidly advancing technology. The overall message of the lesson is that AI can be a powerful tool for learning and productivity, but wisdom, truth, morality, and human dignity ultimately come from human persons and not from machines.

4. AI in Healthcare

A patient was sitting in a dentist chair not too long ago. It was their annual checkup which meant a set of x-rays. After each x-ray, an image appeared on a monitor just to the left of the patient. The image was annotated with color-coding and labels. These annotations and labels were generated by AI. It showed potential trouble spots based on its analysis of the image against a baseline of thousands of images like it.

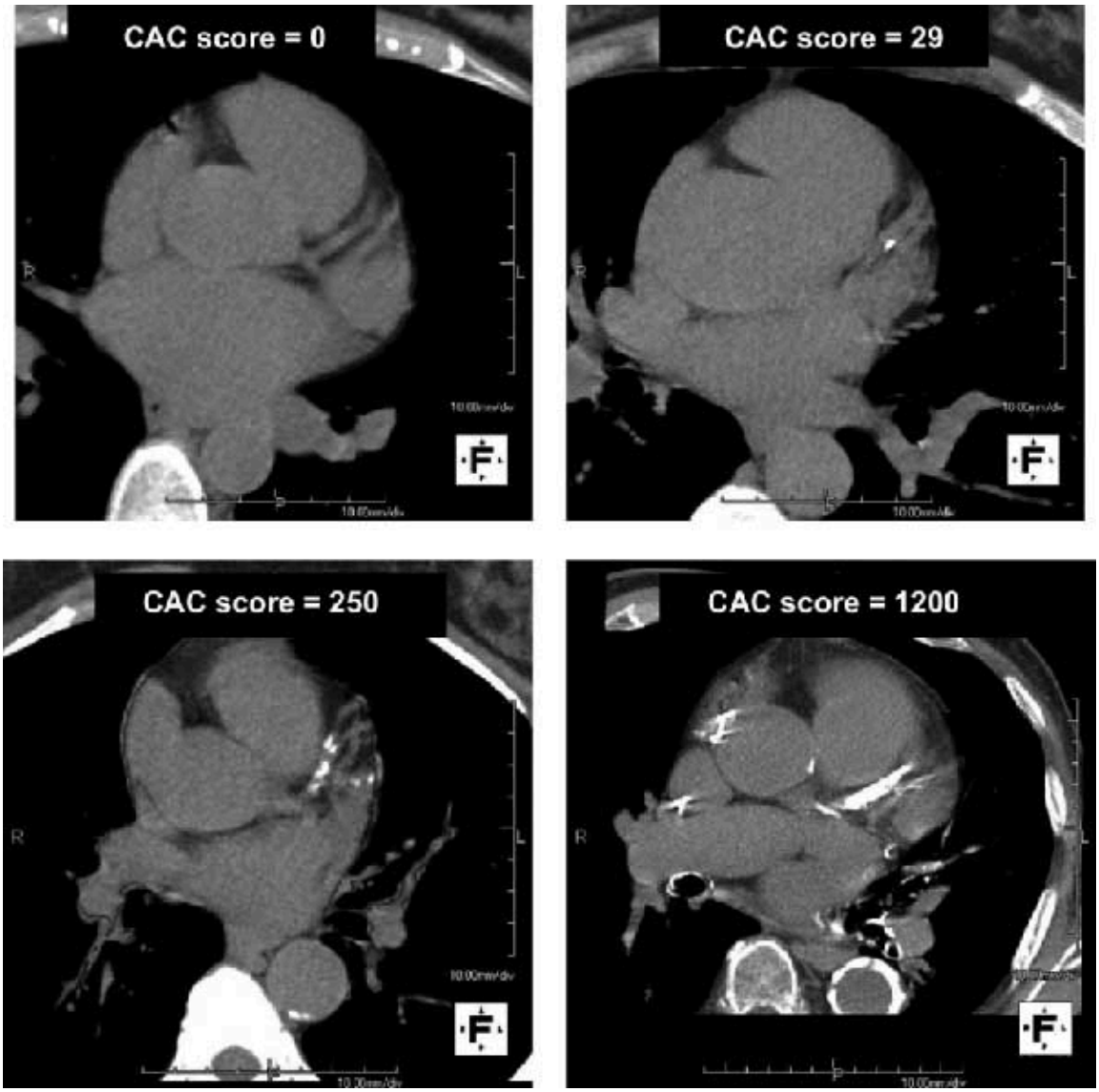


`Total kidney volume` is a measure that attempts to discover kidney issues in patients. It involves analyzing dozens of kidney images, one slide after another — a laborious process that can take about 45 minutes per patient. With recent innovations, researchers now use artificial intelligence (AI) to automate the process, generating results in a matter of seconds across dozens of images.



`LDL` stands for **low-density lipoprotein**, also known as "bad" cholesterol. It is often called “bad” cholesterol because higher levels are strongly associated with atherosclerosis (narrowing and hardening of arteries) and cardiovascular events like heart attacks. A coronary calcium score (CAC) is a means of measuring calcified plaque, a sign of narrowing and hardening of the arteries.

To obtain a CAC, a patient lies on a table while a CT (computed tomography) scan is performed. The set of images contained are compared against an AI model that contains thousands of images along with their scores to calculate a score for the patient. The calcium score determines the level of risk for a patient to have a heart attack or a stroke in following five or 10 years if there was no intervention. It is from this score that a doctor can design a treatment plan to reduce the risk of a coronary event.



When a healthcare professional must complete tasks like writing clinical notes or filling out forms, AI can complete the task faster than traditional methods. In medical records software, professionals begin to enter their notes while AI examines the input. The software begins to propose how to complete the entry by analyzing the current patient's health history. It can

then pre-fill other fields in the form. The professional must then analyze the results and approve the entries before saving the record.

Finally, the use of AI chatbots as informal counselors has been surprisingly widespread. A generation of young are living stressful lives, beset by social media and peer pressure. A generation of the elderly is going through the loneliness of extended lifetimes. There are situations where AI mediation can help in high-stress circumstances. The Sheba Medical Center in Israel has developed a suite of mental health AI support technologies labeled as LIV. For instance, an AI avatar has been found to improve outcomes for those suffering from post-traumatic stress disorder (PTSD).

In the case of all current AI applications and technology, healthcare professionals are still needed to provide:

- Clinical context for the algorithms that train AI.
- Accurate and relevant information for AI to analyze.
- Translation of AI findings to be meaningful for patients

5. AI in Finance

How AI Is Used in Finance

Artificial intelligence is widely used in modern finance to help institutions make faster, more accurate, and more efficient decisions. Finance involves enormous amounts of data: prices, transactions, customer records, loan histories, market news, and economic indicators. Because AI systems are designed to detect patterns in large datasets, they are especially useful in banking, investing, insurance, and financial regulation.

One major use of AI in finance is **fraud detection**. Banks and credit card companies process millions of transactions every day. AI systems can learn what normal activity looks like for a customer and then flag unusual behavior, such as a sudden purchase in another country or a rapid series of large withdrawals. This helps protect customers from theft and helps financial institutions reduce losses. Unlike older rule-based systems, AI can adapt as criminals change their methods.

AI is also used in **credit scoring and lending**. Lenders want to know whether a borrower is likely to repay a loan. AI can analyze financial history, income patterns, debt levels, and other data to estimate risk. This can make lending decisions faster and, in some cases, expand access to credit.

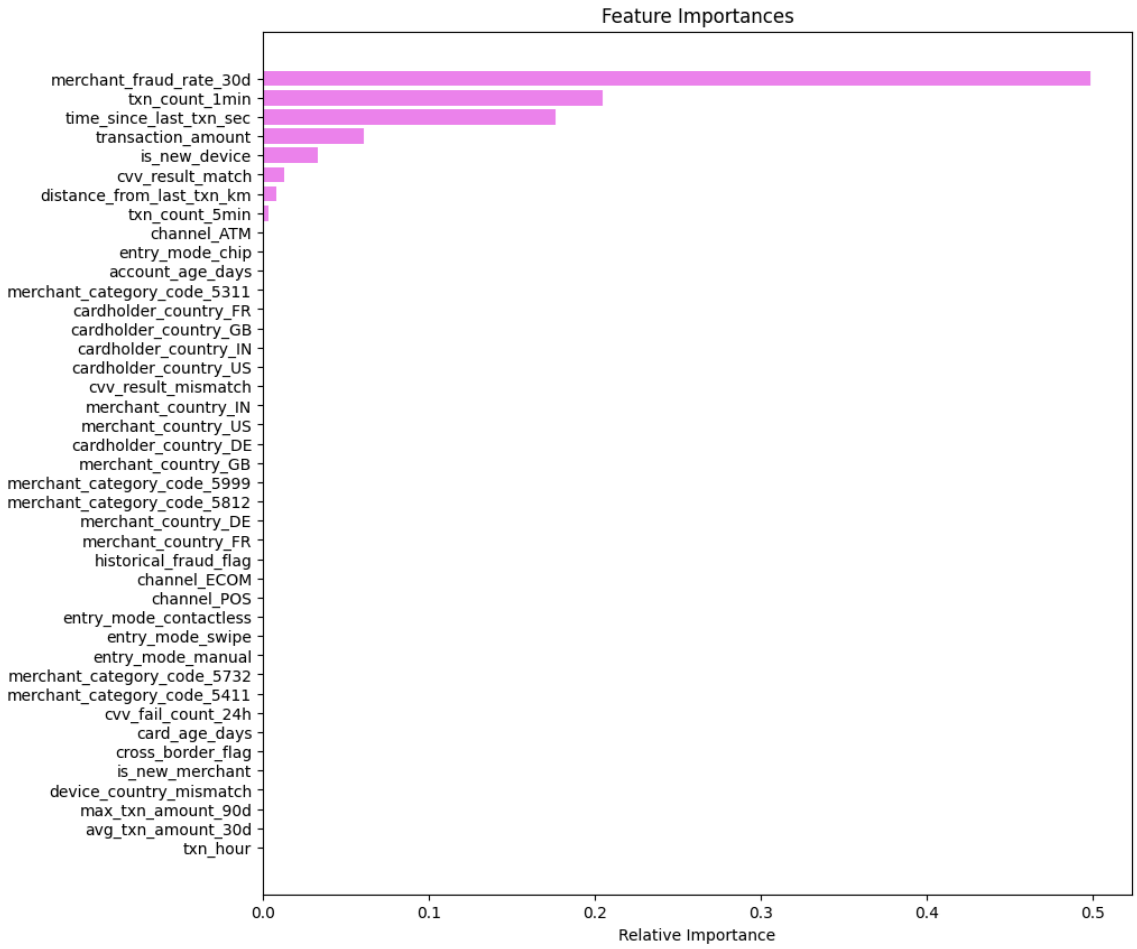
Example:

The business engaged you as a Data Scientist to design and implement a behaviour-driven credit card fraud detection solution that improves real-time fraud identification while

minimizing false declines. The key challenges involve distinguishing between subtle fraudulent behavior and legitimate customer activity, reducing false positives caused by static rules, and ensuring that fraud decisions are explainable and operationally feasible. The expected outcome is to build a scalable and interpretable fraud detection framework using anomaly detection and machine learning models that enables risk-based actions such as transaction approval, step-up verification, or blocking, thereby reducing fraud losses and improving customer experience.

Here are the results of experimenting with different classification training algorithms:

	Decision Tree (Baseline Model)	Decision Tree with class_weight	Decision Tree (Tuned)	Random Forest (Baseline Model)	Random Forest (Tuned)
Accuracy	0.957833	0.962083	0.966917	0.978917	0.984083
Recall	0.658582	0.660448	0.664179	0.656716	0.660448
Precision	0.522189	0.564593	0.621291	0.836105	0.975207
F1	0.582508	0.608770	0.642020	0.735632	0.787542



Another important area is **investment management**. AI tools can analyze market trends, company performance, news reports, and economic data to help investors make decisions.

Some investment firms use AI-powered trading systems that buy and sell financial assets automatically. These systems can react in fractions of a second, much faster than human traders. While this can improve efficiency, it can also increase market volatility if many automated systems respond to the same signals at once.

AI is also used in **customer service**. Many banks now use chatbots and virtual assistants to answer common questions, help customers check balances, explain transactions, or guide users through online services. These tools can provide support at any time of day. Still, human assistance remains important, especially when people face complex problems, financial stress, or personal hardship.

In **insurance**, AI helps companies assess risk, set premiums, process claims, and detect false claims. For example, AI may analyze driving habits for auto insurance or evaluate property damage from photographs. These uses can make insurance more efficient, but they must be handled carefully to avoid invading privacy or treating people as merely data points.

Catholic social teaching offers principles to evaluate and guide AI in finance:

- Human dignity: financial systems should respect persons as ends, not merely data points. Decisions that affect access to housing, credit, or employment require human oversight and recourse.
- Preferential option for the poor: AI should reduce, not entrench, economic exclusion. Models that extend responsible credit or financial education to the poor align with solidarity.
- Solidarity and the common good: firms should assess systemic impacts of algorithms and prioritize stability, fairness, and public trust over short-term profit.
- Stewardship and prudence: institutions must responsibly manage data and computational power, ensuring safety, explainability, and auditability.
- Adopt human-in-the-loop review for critical decisions; require algorithmic impact assessments; enforce explainability and appeals processes; run bias audits and use diverse training data; protect privacy and obtain meaningful consent; regulate to mitigate systemic risks and concentration.

6. AI in Agriculture

Examples:

- Crop monitoring using drones
- Soil analysis
- Smart irrigation systems
- Detecting plant disease

- Autonomous tractors

How AI Is Used in Agriculture

Artificial intelligence helps farmers grow food more efficiently, manage resources wisely, and respond to environmental challenges. Agriculture has always depended on observation: farmers watch the soil, weather, animals, pests, and crops. AI expands this ability by analyzing large amounts of data from sensors, satellites, drones, weather systems, and farm equipment. In this way, AI can help farmers make better decisions about planting, watering, fertilizing, harvesting, and caring for livestock.

One important use of AI in agriculture is **precision farming**. Instead of treating an entire field the same way, AI can help identify the specific needs of different areas. For example, one part of a field may need more water, while another may need less fertilizer. Sensors and satellite images can collect information about soil moisture, crop health, and nutrient levels. AI systems analyze this data and recommend targeted action. This can reduce waste, lower costs, and protect the environment by limiting overuse of water, pesticides, and fertilizers.

AI is also used for **crop monitoring and disease detection**. Drones and cameras can take images of plants, and AI systems can identify signs of disease, pests, or nutrient problems earlier than the human eye might notice. Early detection allows farmers to act before a problem spreads across an entire field. This can help protect food supplies and reduce the need for broad chemical treatments.

Another major use is **weather and yield prediction**. Farming depends heavily on weather, and uncertainty can create serious risks. AI can analyze weather forecasts, historical climate patterns, soil data, and crop information to estimate likely yields or warn farmers about drought, flooding, frost, or heat stress. These predictions can help farmers plan planting times, irrigation, harvesting, storage, and market decisions.

AI is also used in **automated machinery and robotics**. Some tractors, harvesters, and irrigation systems can operate with limited human control. Robots may help pick fruit, remove weeds, or monitor livestock. In dairy farming, AI can support robotic milking systems and monitor animal health by tracking movement, feeding, and behavior patterns. These tools can reduce repetitive labor and help farmers identify problems quickly.

Catholic ethics reminds us that technology should serve people, not replace human responsibility. Agriculture is not merely an industry; it is a vocation connected to food, land, family, community, and creation. AI should therefore be used in ways that respect farmers, protect workers, care for animals, preserve the environment, and promote food security. The central question is not only whether AI can make farming more productive, but whether it helps agriculture better serve life, justice, and the common good.

Check out this video about how AI can weed a field using lasers instead of using herbicides.

Have a look at self-driving tractors.

7. AI in Manufacturing & Transportation

When we think of artificial intelligence, it is easy to imagine chatbots or virtual assistants living inside our screens. But AI is also hard at work in the physical world, making the things we use every day and getting them to our doorsteps. In the interconnected realms of manufacturing and transportation, AI acts as a brilliant, untiring coordinator.

AI in Manufacturing: The Smart Factory In modern manufacturing, AI transforms how goods are designed, created, and inspected.

First, AI excels at **predictive maintenance**. A factory is full of expensive, complex machinery. By analyzing data from sensors that monitor temperature, sound, and vibration, AI can predict exactly when a machine is likely to fail. Before a robotic arm decides to take an unplanned vacation, the AI alerts technicians to replace a worn-out part. This keeps factory floors running smoothly and protects human workers from sudden, dangerous equipment failures.

Second, AI is a master of **quality control**. Through "computer vision," AI systems inspect items on fast-moving assembly lines. Whether it is checking a microchip for microscopic flaws or ensuring a jar of baby food is perfectly sealed, AI can spot defects with superhuman speed and accuracy, ensuring safer products reach consumers.

AI in Transportation: Moving the World Once a product is made, it has to go somewhere. Transportation and logistics rely heavily on AI to move people and goods safely and efficiently.

The most common use is **route optimization**. For delivery trucks, cargo ships, and airplanes, AI analyzes traffic, weather patterns, and historical data to chart the fastest, most fuel-efficient paths. If a storm is brewing over the Atlantic, an AI system can instantly recalculate a ship's route to avoid it.

AI is also the driving force behind **vehicle safety and autonomy**. The automated safety features in modern cars—like emergency braking and lane-keeping assist—rely on AI to react to sudden dangers faster than a human foot can hit the brake pedal. While fully autonomous, self-driving cars are still being refined, the ultimate goal is to drastically reduce accidents caused by human error.

A Catholic Perspective: Dignity, Creation, and the Common Good These physical applications of AI provide a wonderful opportunity to reflect on our moral priorities.

First, we must consider the **dignity of work**. Catholic social teaching insists that work is for the person, not the person for work. Ideally, AI in a factory takes over dull, dirty, and dangerous tasks, freeing human beings for safer work that requires creativity and complex

problem-solving. However, we must ensure that automation does not discard workers or treat them as mere cogs in a machine. Society has a duty to help retrain workers displaced by new technologies.

Second, AI offers powerful tools for **care for creation**. Pope Francis's encyclical *Laudato Si'* calls us to protect our common home. When AI optimizes shipping routes to save millions of gallons of fuel, or reduces material waste on a factory floor, it acts as an instrument of environmental stewardship.

Ultimately, AI in manufacturing and transportation should be measured by how well it serves the **common good**. When used ethically, these technologies make our roads safer, our work more dignified, and our use of earth's resources more sustainable.

8. Exit Ticket

1. Name one way AI is used in healthcare and explain how it helps patients or doctors.
2. Why are human doctors and healthcare workers still needed even when AI systems are very accurate?
3. What is fraud detection in finance, and how can AI help with it?
4. Give one example of how AI can improve farming or agriculture.
5. What is predictive maintenance in manufacturing?

9. Reflection

Would you personally trust AI to help make decisions in healthcare or finance? Why or why not?

10. Vocabulary

1. Predictive Analytics: The use of AI to analyze data and predict future outcomes.

   Example: In healthcare, AI analyzes thousands of kidney images to quickly predict potential kidney problems.

2. Fraud Detection: AI systems that identify unusual or suspicious activities to prevent theft or fraud.

   Example: Banks use AI to spot strange credit card transactions, like sudden purchases in another country.

3. Medical Diagnosis: Using AI to help doctors identify diseases or health issues by analyzing medical data.

   Example: AI analyzes dental x-rays to highlight possible trouble spots for dentists.

4. Automation: The process where AI performs tasks without human intervention, often to save time or reduce errors.

   Example: AI automates the analysis of dozens of kidney images in seconds instead of 45 minutes.

5. Computer Vision: AI technology that enables machines to "see" and interpret images or videos.

   Example: In manufacturing, AI inspects products on assembly lines to find defects.

6. Recommendation System: AI tools that suggest products, services, or actions based on data patterns.

   Example: Financial chatbots recommend next steps to customers based on their transaction history.

7. Decision Support System: AI systems that assist humans in making better decisions by providing analyzed data and suggestions.

   Example: AI helps doctors by pre-filling medical forms based on patient history, but the doctor reviews and approves the entries.

8. Algorithmic Bias: When AI systems unfairly favor or discriminate against certain groups due to biased training data.

   Example: AI in finance must be carefully checked to avoid unfairly denying loans to certain people.

9. Efficiency: How AI helps complete tasks faster, more accurately, or with less waste.

   Example: AI optimizes delivery routes to save fuel and time in transportation.

10. Human Oversight: The role of people in supervising and approving AI decisions to ensure accuracy and ethics.

    Example: Doctors review AI medical diagnoses before making treatment decisions.

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